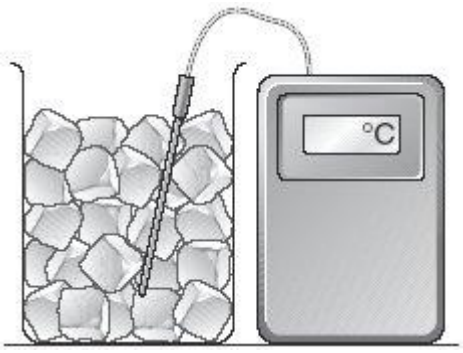
Q1. (a) Draw a line from each change of state to the correct name.
Draw only **four** lines.

change of state	name
solid to liquid	evaporating
liquid to gas	melting
gas to liquid	condensing
liquid to solid	freezing

3 marks

(b) Kate made some ice cubes from pure water.
She used a sensor to measure the temperature of the ice.



What temperature will the sensor show when the ice is melting?

..... °C

1 mark

(c) Kate made some more ice cubes from salt solutions. She used a different amount of salt in each ice cube.

The table shows the temperature at which the ice cubes melted.

mass of salt in each ice cube (g)	temperature ice cube melted (°C)
5	−4
10	−8
15	−11
20	−15

Look at the table above.
As the mass of salt increased, what happened to the temperature at which the ice cube melted?

.....

1 mark

(d) In very cold weather a mixture of salt and sand is spread on roads.

Why are salt **and** sand used?
Tick the **two** correct boxes.

Salt makes the roads white.	☐	Sand dissolves in water.	☐
Salt makes water freeze.	☐	Sand increases friction between car tyres and the road.	☐
Salt makes ice melt.	☐	Sand makes water freeze.	☐

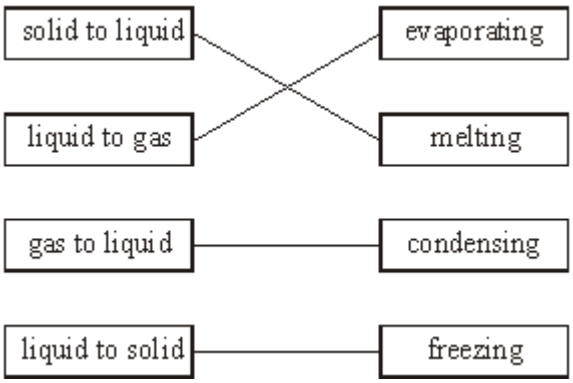
2 marks

maximum 7 marks

Mark schemes

Q1.

(a) •



*award three marks for **all four** correct lines*
*award two marks for any **three** correct lines*
*award one mark for any **two** correct lines*
if more than one line is drawn from any change of state, do not credit that change of state

3 (L3)

(b) • 0 °*C*

accept 'zero'
*do **not** accept 'nothing'*

1 (L4)

(c) • it decreased

accept 'it got colder'
'it dropped to below 0°C' is insufficient
any references to time are insufficient

1 (L3)

(d) • Sand increases friction between car tyres and the road. ✓

• Salt makes ice melt. ✓

if more than two boxes are ticked, deduct a mark for each incorrect box
minimum of zero

2 (L4)

[7]